

WD Red Pro WD103KFBX

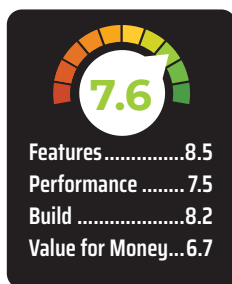
A fast, sturdy NAS drive for the pros

The WD Red Pro WD103KFBX is a very specialised hard drive. While it's easy to assume that hard drives are commoditised and barely stand apart from each other, the truth is that the firmware that sits at the heart of these drives have been fine-tuned for the applications that they're meant to be used. The WD Red Pro, for example, is meant for round-the-clock usage with negligible downtime. So these are the ones that go into NAS (Network Attached Storage) devices that need to maintain significant uptime. Unfortunately, it's not the cheapest way to add 10 TB of stage given its price tag of Rs 49,599. It's basically meant for folks who want reliability in a heavy-use scenario.

SPECIFICATIONS

At first glance, the WD103KFBX has the right ingredients for heavy use. It uses CMR recording, spins at 7200 RPM, carries a large 512 MB cache and is rated for NAS environments rather than occasional desktop use. A lot of NAS drives used to be 5400 RPM drives but of late, we're seeing more of 7200 RPM drives being pushed into the segment. That's great for performance but the main trade-off is acoustics. This is a performance-oriented mechanical drive, and under random access, it sounds like one.

Aside from being a 10TB 3.5-inch SATA hard drive, the WD Red Pro WD103KFBX uses conventional magnetic recording, which is important for those seeking performance in multi-drive storage because CMR drives generally handle sustained writes and RAID rebuild behaviour more predictably than SMR-based drives. WD rates this 10 TB model for up to 267 MB/s transfer



speeds, and the Red Pro family is also positioned for RAID-optimised NAS systems. Surprisingly, this is an air-filled drive.

SETUP AND INITIALISATION

For this review, two WD Red Pro WD103KFBX 10TB drives were installed in a QNAP TS-216G NAS and configured in RAID 1. We went for this test environment because it reflects how many home-office users, creators and small businesses are likely to deploy a pair of high-capacity NAS drives: mirrored for redundancy, connected over 2.5 GbE and used for media libraries, backups, shared folders and light application workloads.

Installation in the QNAP TS-216G was straightforward. Once the two WD Red Pro drives were added and the storage pool was created, QTS reported healthy S.M.A.R.T. status for both drives.

There were no reallocated sectors, no spin retry count warnings and the power-on hours matched the testing.

The initial RAID 1 synchronisation took approximately 14 to 18 hours. That sounds long, but it is expected when the NAS has to initialise and mirror a high-capacity pair of mechanical drives. The important bit is that the NAS remained usable during the process. File access, the QTS interface and basic app behaviour continued to work, although response times were slightly higher while the background sync was active.

This is one of the areas where NAS drive behaviour matters. A weaker desktop-class drive may technically work inside a NAS, but long background operations such as initialisation, rebuilds and indexing are precisely the sort of tasks that separate purpose-built NAS drives from general storage. The WD Red Pro drives did not throw up any health warnings or some obvious responsiveness issues during this stage.

SYNTHETIC PERFORMANCE

The strongest numbers came from sequential testing. Inside QNAP's Storage & Snapshots benchmark, the individual drives posted sequential read and write speeds in the 260 MB/s to 270 MB/s range. That is very close to the official rated ceiling for this model and is strong for a mechanical hard drive. Over the network, the results were even more telling. CrystalDiskMark runs over the TS-216G's 2.5GbE connection produced stable read and write speeds between 280 MB/s and 295 MB/s for both 1 GB and 16 GB test sizes. That effectively means the NAS network link became the limiting factor rather than the drives. In a two-bay RAID 1 setup, the WD Red Pro pair had no difficulty filling the available 2.5 GbE pipe.

On a basic Gigabit Ethernet NAS, these drives would spend most of their time waiting on the network. Gigabit Ethernet typically caps real-world transfers a little above 100MB/s, far below what the WD103KFBX can actually